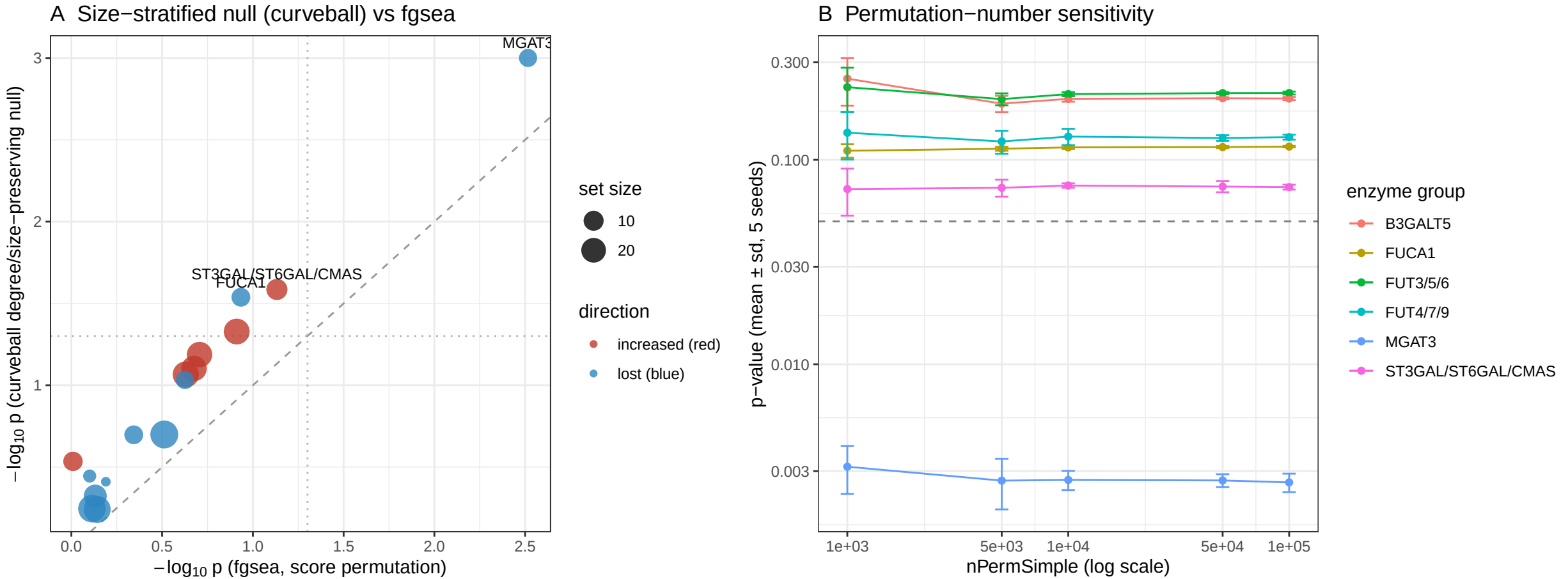
A Size−stratified null (curveball) vs fgsea
MGAT3
ST3GAL/ST6GAL/CMAS
FUCA1
−log10 p (curveball degree/size−preserving null)
−log10 p (fgsea, score permutation)
0.0
0.5
1.0
1.5
2.0
2.5
1
2
3
set size
10
20
direction
increased (red)
lost (blue)
B Permutation−number sensitivity
p−value (mean ± sd, 5 seeds)
0.300
0.100
0.030
0.010
0.003
1e+03
5e+03
1e+04
5e+04
1e+05
nPermSimple (log scale)
enzyme group
B3GALT5
FUCA1
FUT3/5/6
FUT4/7/9
MGAT3
ST3GAL/ST6GAL/CMAS